CSO Lab 1:Electronic

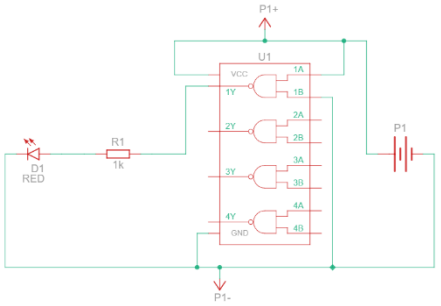
Switch the input according to the truth table below and complete the output column.

Q1:

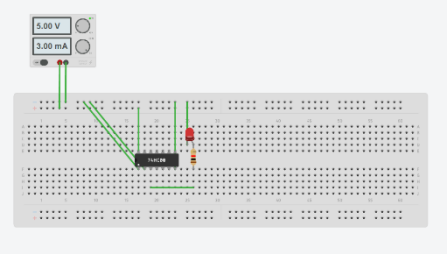
74LS00

Input A	Input B	Output
0	0	1
0	1	1
1	0	1
1	1	0

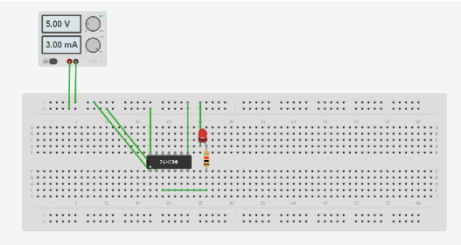
Schematic diagram



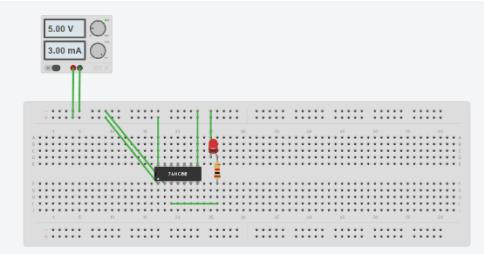
Input: A=0, B=0 Output: 1



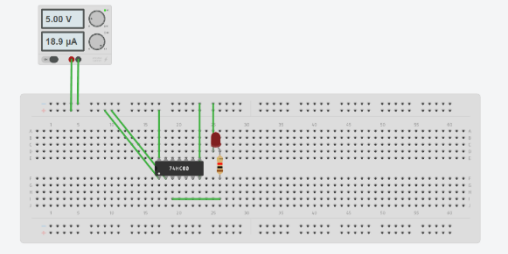
Input: A=0, B=1 Output: 1



Input: A=1, B=0 Output: 1



Input: A=1, B=1 Output: 0

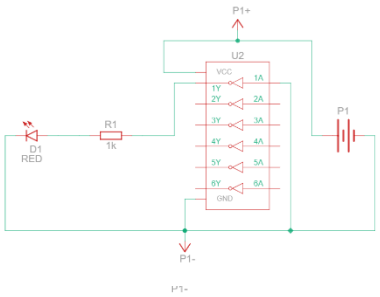


Q2:

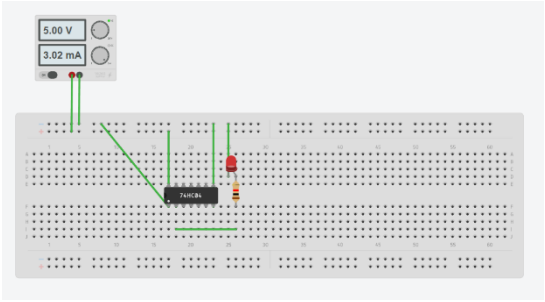
74LS04

Input	Output
0	1
1	0

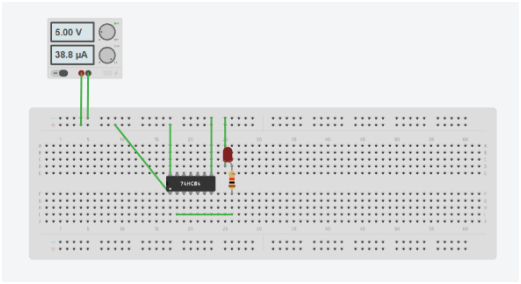
Schematic diagram



Input: A=0 Output: 1



Input: A=1 Output: 0

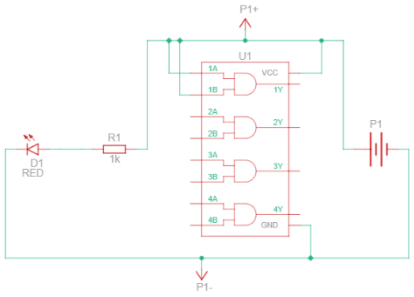


Q3:

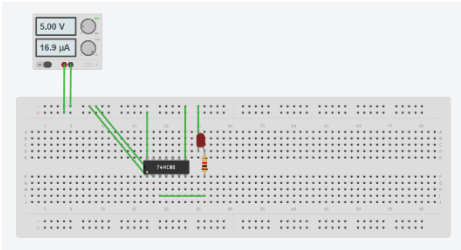
74LS08

Input A	Input B	Output
0	0	0
0	1	0
1	0	0
1	1	1

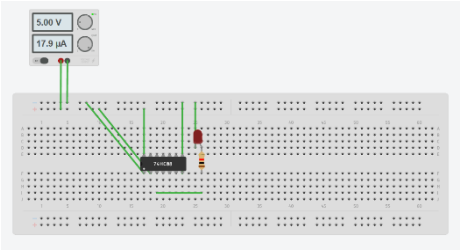
Schematic diagram



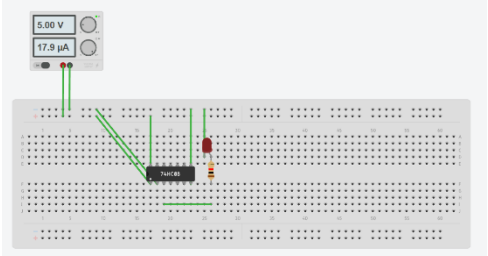
Input: A=0, B=0 Output: 0



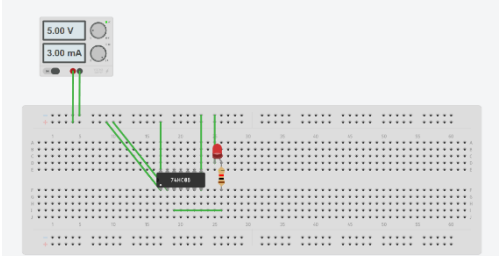
Input: A=0, B=1 Output: 0



Input: A=1, B=0 Output: 0



Input: A=1, B=1 Output: 1

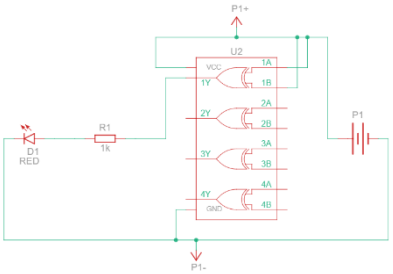


Q4:

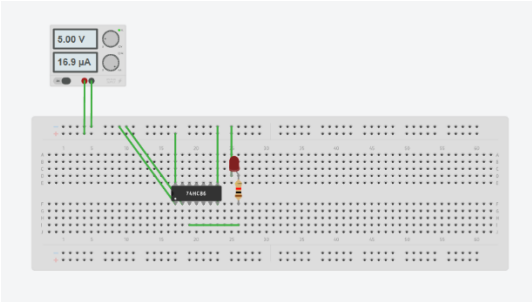
74LS86

Input A	Input B	Output
0	0	**0**
0	1	**1**
1	0	**1**
1	1	**0**

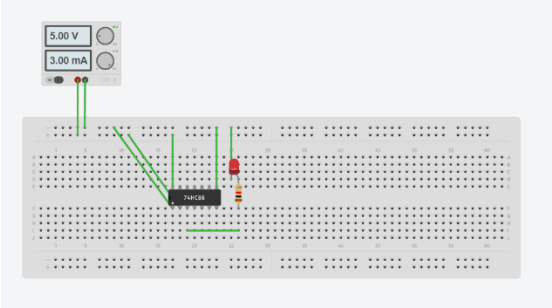
Schematic diagram



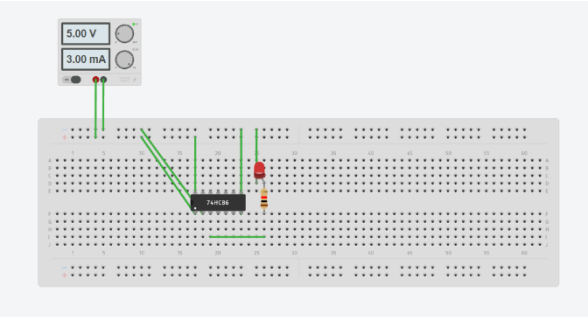
Input: A=0, B=0 Output: 0



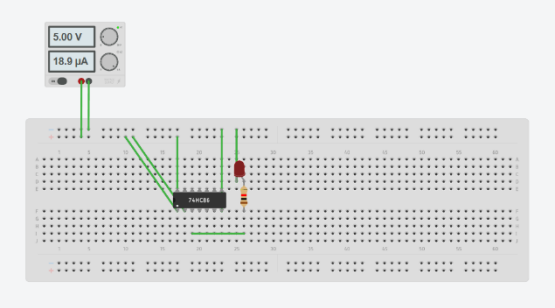
Input: A=0, B=1 Output: 1



Input: A=1, B=0 Output: 1



Input: A=1, B=1 Output: 0

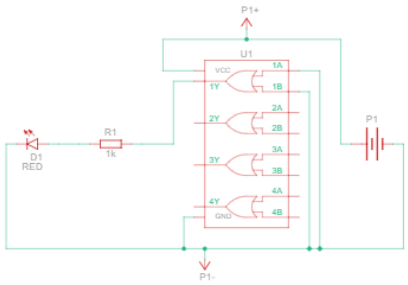


Q5:

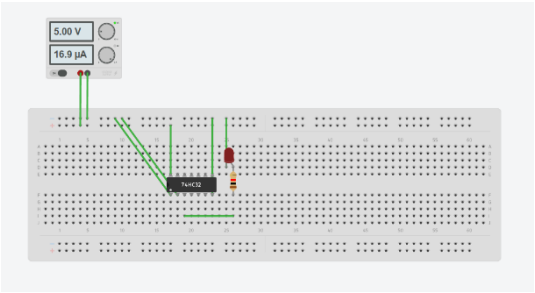
74LS32

Input A	Input B	Output
0	0	**0**
0	1	**1**
1	0	**1**
1	1	**1**

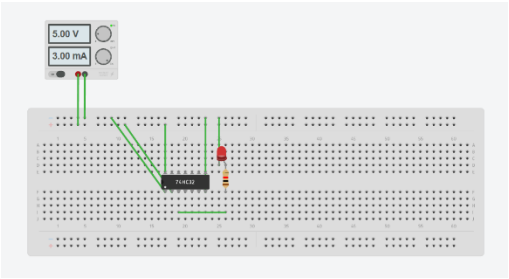
Schematic diagram



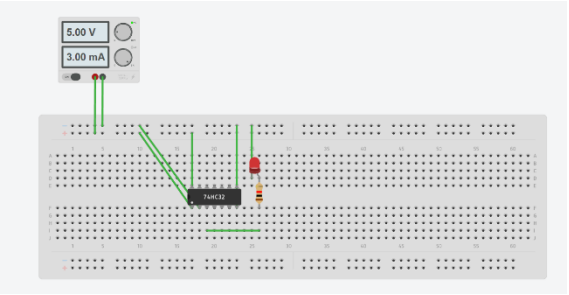
Input: A=0, B=0 Output: 0



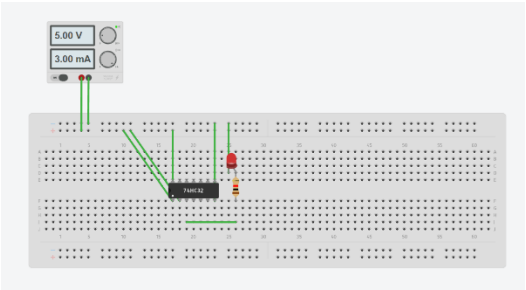
Input: A=0, B=1 Output: 1



Input: A=1, B=0 Output: 1



Input: A=1, B=1 Output: 1



Result:

Based on the truth table value obtained by building a circuit, we can conclude that:
74LS00 is NAND gate
74LS04 is NOT gate
74LS08 is AND gate
74LS86 is XOR gate
74LS32 is OR gate